

Predicting Crash Severity: Analysis Overview

Portfolio project overview

Cleaned version with
personal and team identifiers
removed.



Background

The Australian Department of Transport has highlighted that the fatality rates across both sexes have seen a consistent drop from 2008 - 2022 (Department of Infrastructure, Transport, Regional Development, Communications and the Arts, 2024)

Domain Problem

Predict crash severity to inform road safety policies (Target high-risk driver types) (Thompson et al., 2018)

Data Science Problem

Binary Classification (Fatal Vs Non-Fatal) and Multiclass classification (5 Severity Levels) using supervised learning (Soltani et al., 2025)

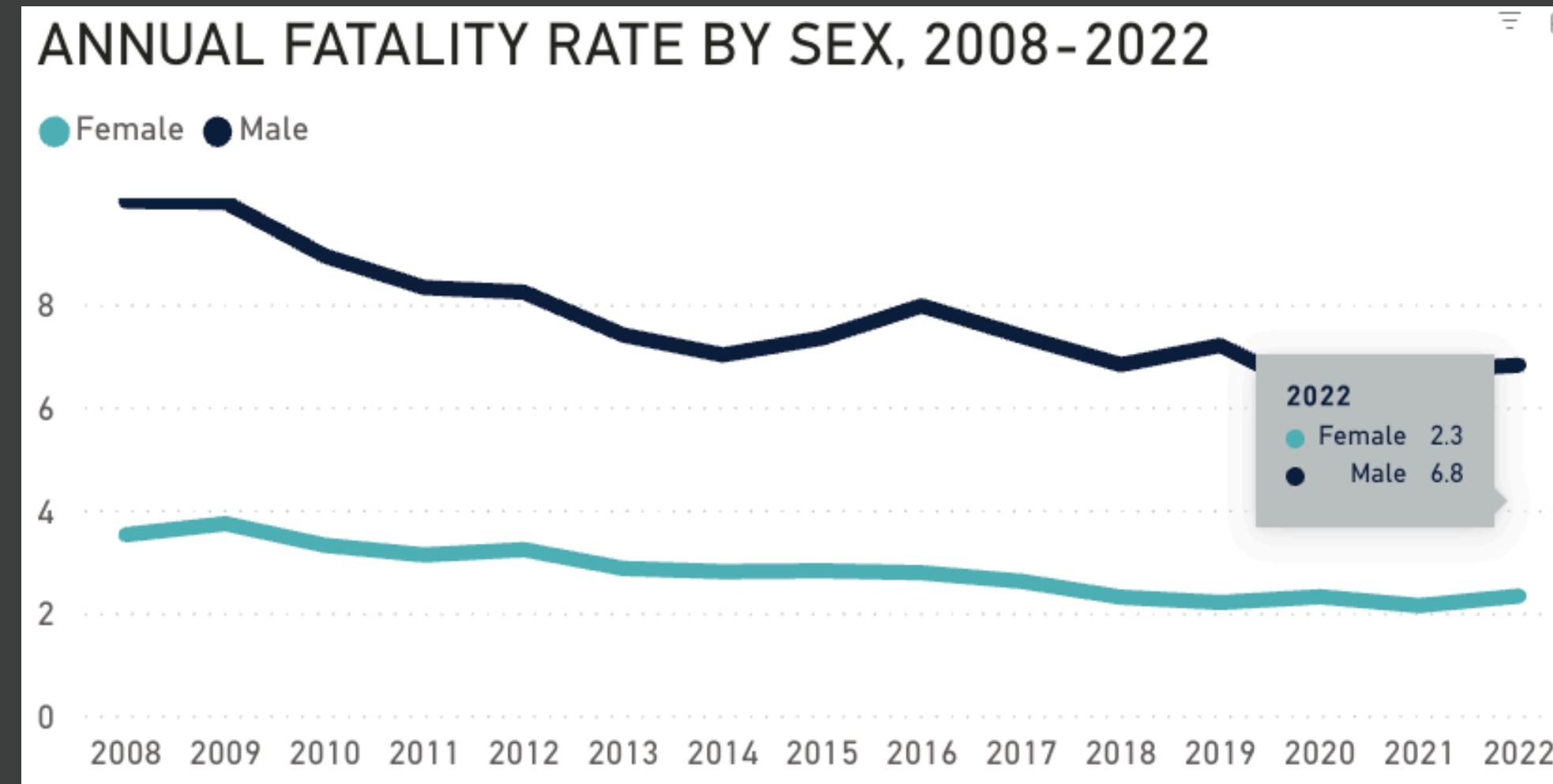


Figure 1: National Crash Dashboard (2008 - 2022)



Dataset

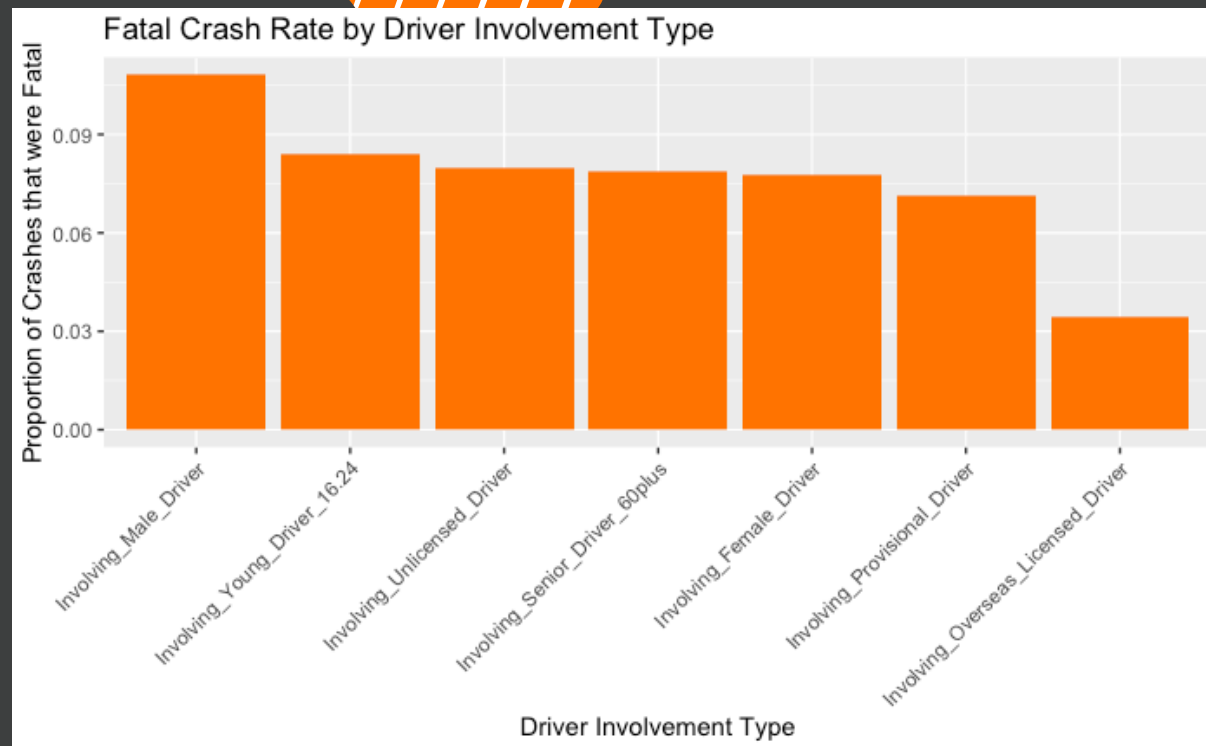


Figure 2: Fatal Crash Proportion by Driver Involvement Type

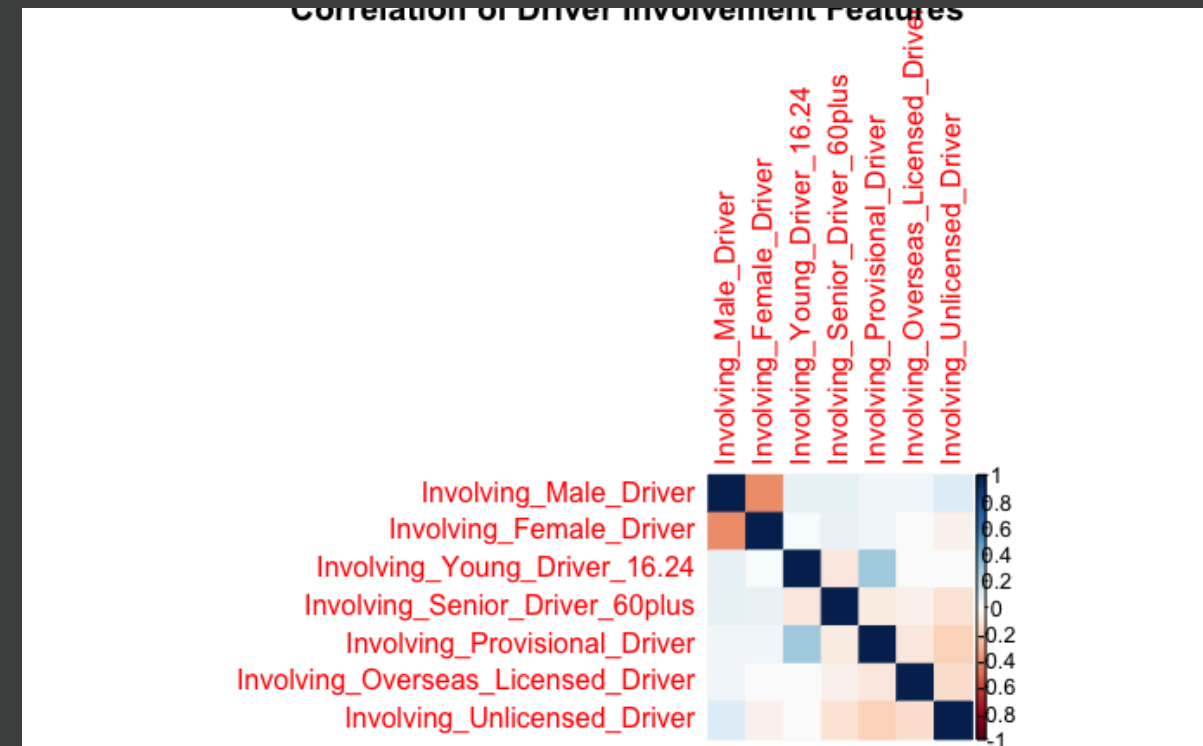


Figure 3: Correlation of Driver Involvement Features

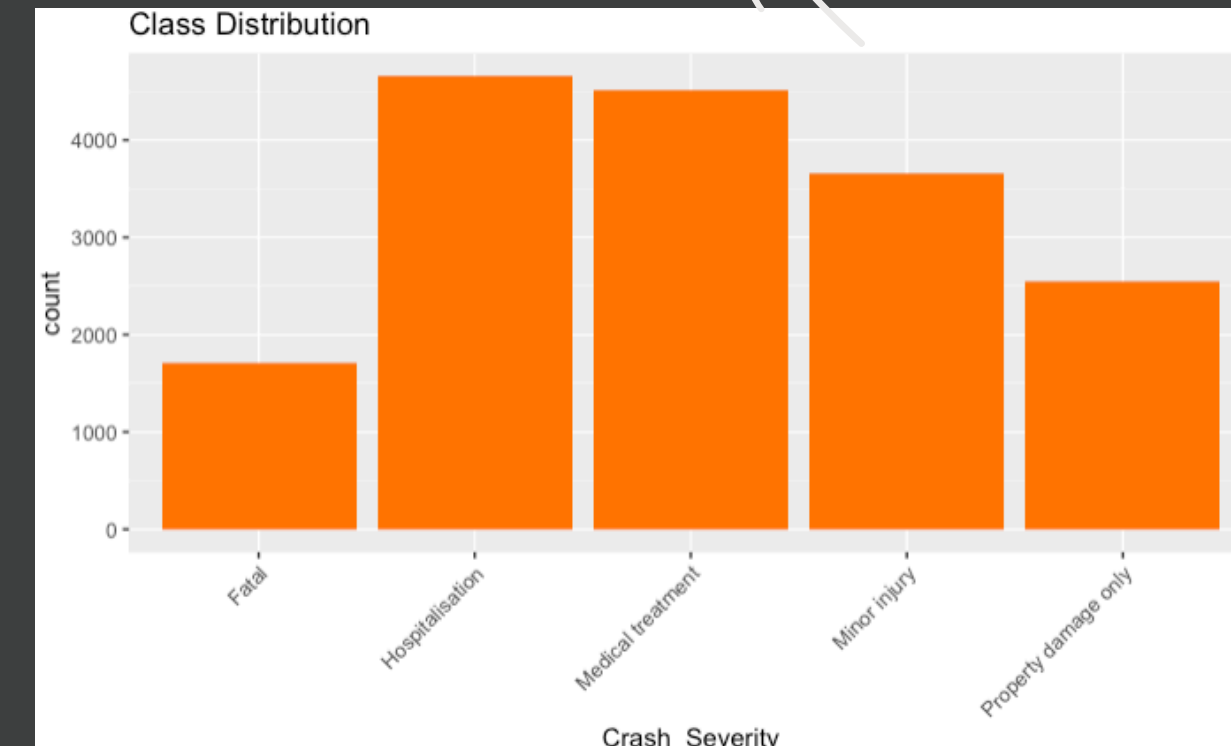


Figure 4: Class Distribution by Crash Severity

Source

- Australian Government Dataset
- 21,350 Entries
- From 2001 - 2023
- Categorises Crash Severity according to Year, Region and Driver Characteristics

Insights

- Male Drivers have highest fatality rate
- Class distribution is imbalanced whereby Fatality rates are almost twice as rare as non-fatal outcomes
- Weak Correlations between features except Male and Female Driver

Inference

- Class Imbalance will have to be addressed (Li et al., 2023)
- Dataset has limited set of features (Only 7)
- The Features are independent as highlighted in the Correlation plot

Shared Methods

- Preprocessing
 - Convert Yes/No → 1/0
 - Create FatalBinary (1 = Fatal, 0 = Others)
- Data Splitting
 - 80/20 Train/Test Split
 - Upsampled Fatal class in training data to address Imbalance
- Binary Focus
 - Use binary instead of multiclass due to poor performance

Random Forest

Evaluation Results: Which resampling strategy works best?

Challenge

- Fatal crash class is underrepresented (imbalance)

Solution

- Applied four main strategies before training for Random Forest
 - Downsampling: Remove non-fatal cases
 - Upsampling: Duplicating fatal cases
 - ROSE: Generating synthetic fatal cases
 - SMOTE: Creating synthetic fatal samples by using their nearest neighbours

Key Result

- Upsampling identified more fatal crashes than other models (68% of actual fatal crashes)
 - Results: Sensitivity (Confusion matrix)
 - Discriminative power (ROC/AUC)

Random Forest Comparison by Resampling Strategy

Models	Accuracy (Balanced)	Sensitivity(Fatal)	AUC	Kappa	Outcomes
Downsampling	62.20%	67.84%	0.6988	0.1226	Balanced with moderate overall
Oversampling ROSE	63.60%	67.61%	0.7026	0.1321	Best AUC, stable metrics
Oversampling SMOTE	84.10%	20.66%	0.6838	0.1182	Highest acc, but detection is poor
Upsampling	62.20%	67.84%	0.7014	0.1226	Best sensitivity, proper balance

- Accuracy: Overall correct predictions
- Sensitivity: % of actual fatal crashes correctly detected
- AUC: Model's ability to distinguish between classes

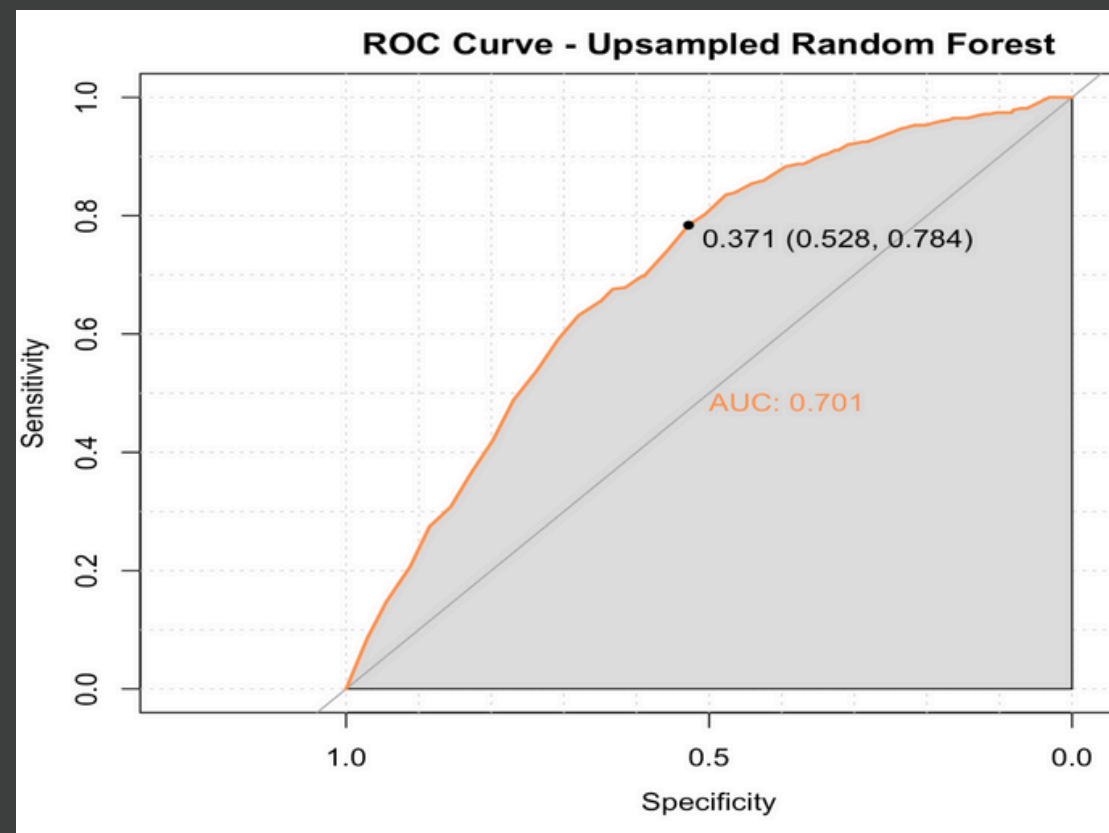


Figure 5: Binary Random Forest ROC Curve

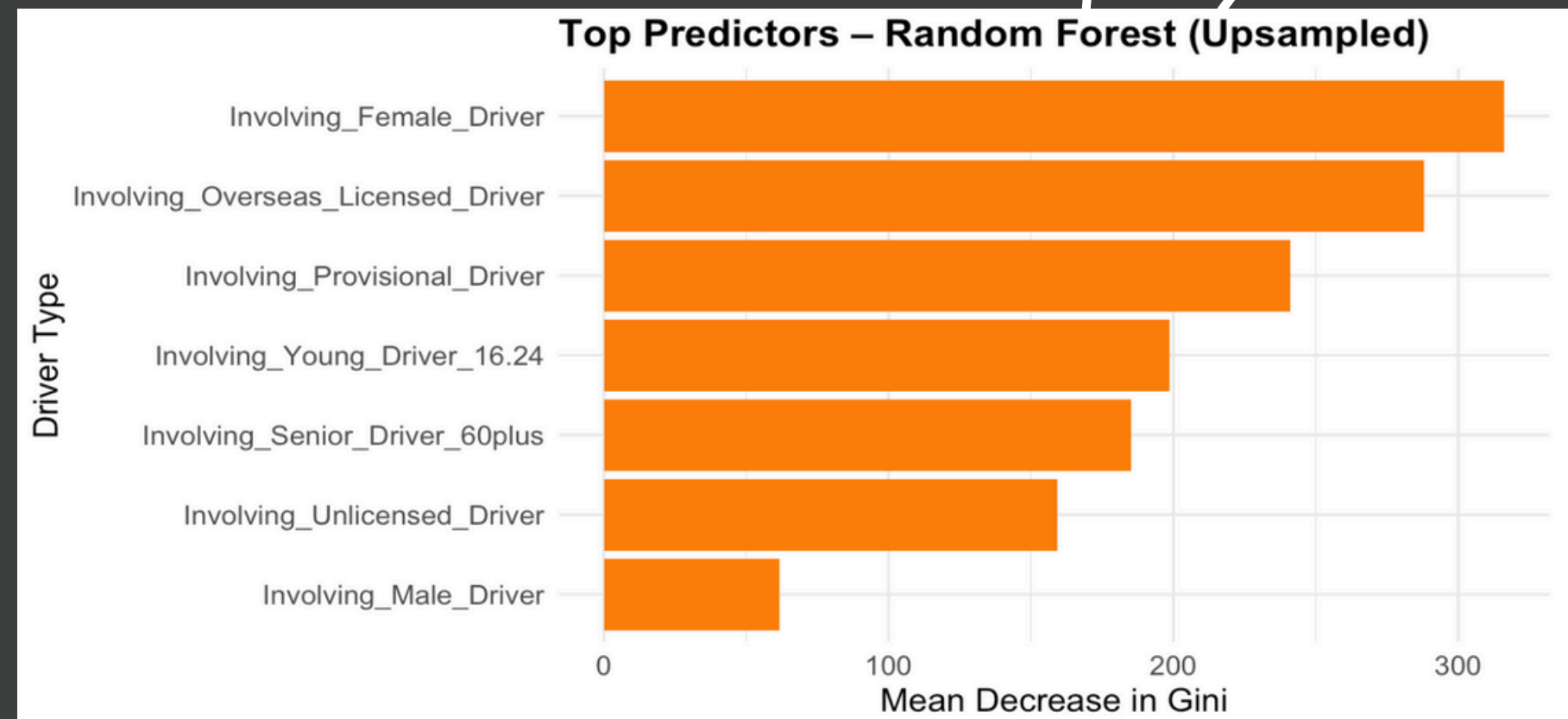


Figure 6: Binary Random Forest Variable Importance

KNN

METHODS - ADDING NOISE

Challenge

- class::knn limitation with tie-breaking for binary features.
 - Identical/very close data points cause "too many ties" error.

Solution

- Added small random noise ($1e-8$).
 - Slightly perturbs data points.
 - Ensures unique distances for KNN.

Result

Model runs without errors.

RESULT - INTERPRETATION - EVALUATION

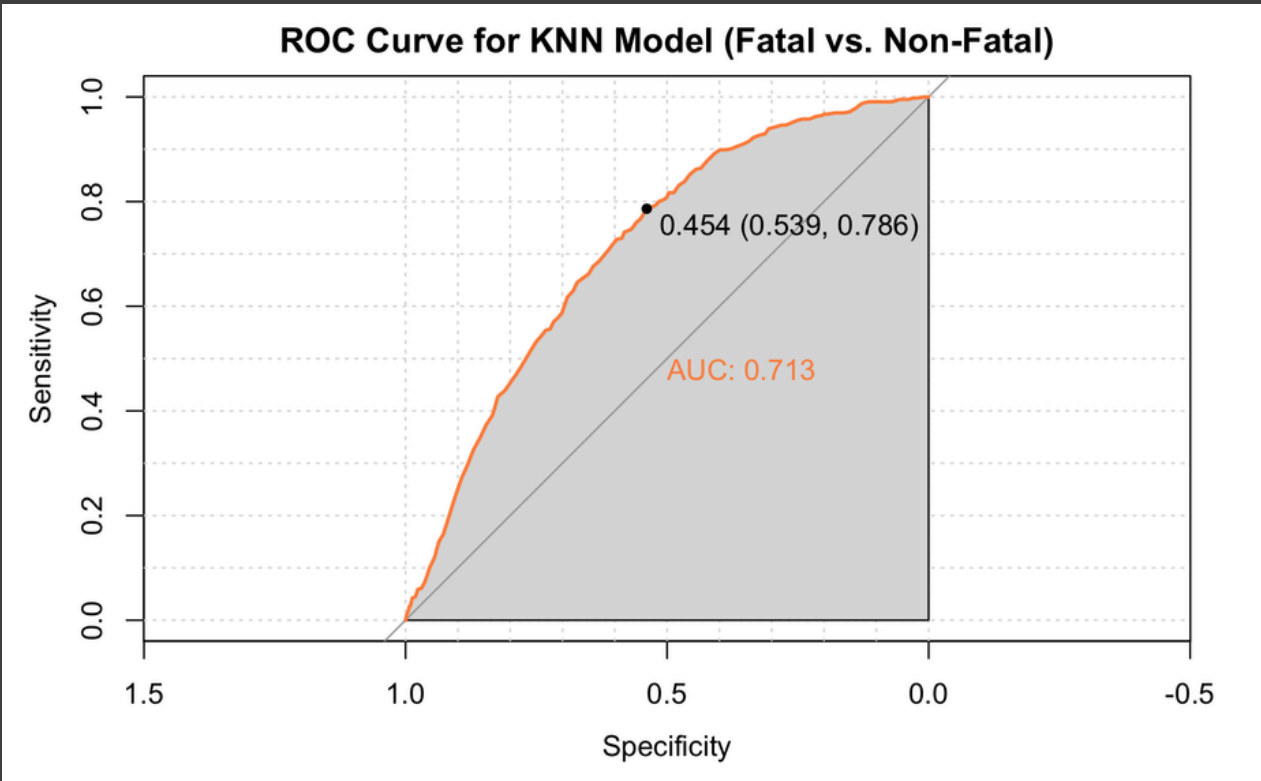


Figure 7: ROC Curve

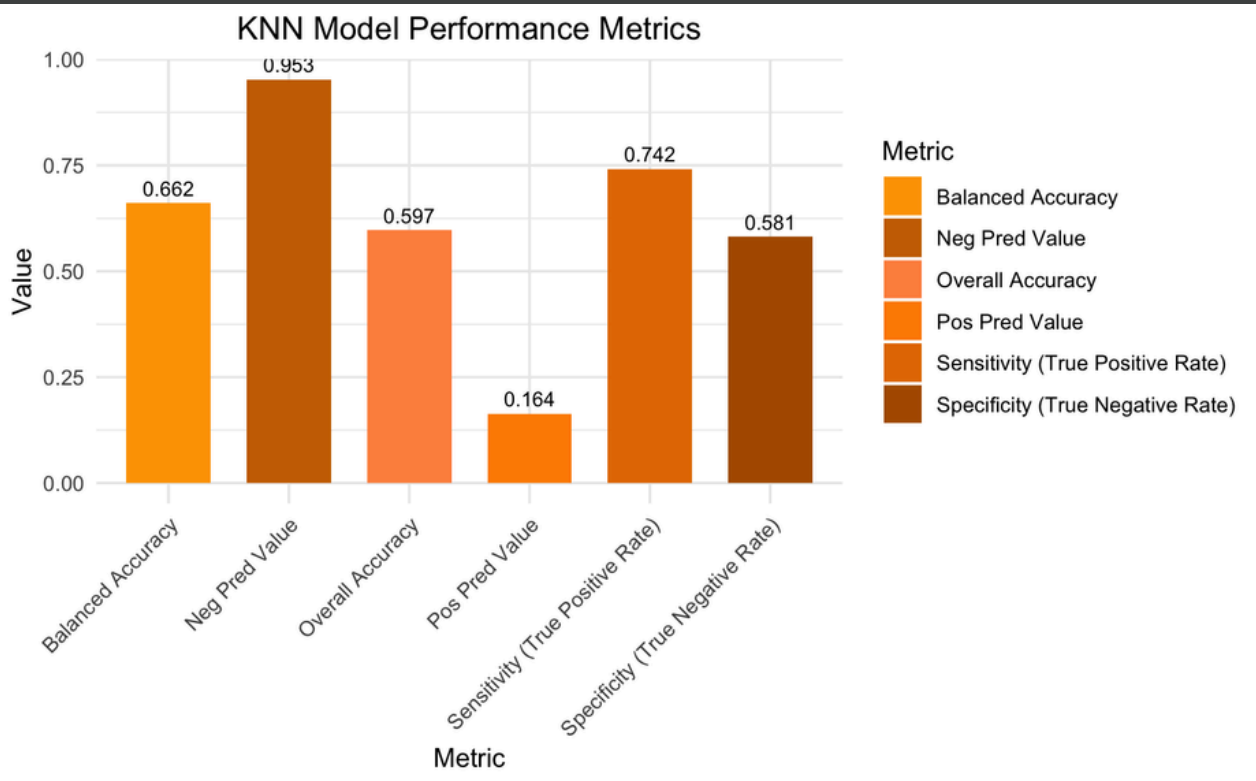


Figure 8: KNN Model Performance

Model Confusion Matrix		
Models	True Non-Fatal	True Fatal
Predicted Non-Fatal	2,236	110
Predicted Fatal	1,607	316

Model Accuracy Metrics	
Balanced Accuracy	66.20%
Sensitivity (Fatal)	74.20%
Kappa	0.126

Logistic Regression

Challenge

- Assumption of linear relationship between predictors and log-odds of outcome.
 - Real-world crash severity likely involves complex, non-linear patterns.
- Feature Selection
 - Risk of overfitting

Solutions

- Used backward propagation to reduce number of predictors
- Fitted full model to compare with output of backward Propagation

Result

- Full model has slightly higher accuracy
 - Full -> 75.3%
 - 3 Features -> 74.1%
 - Female Driver, Provisional Driver and Overseas Licensed Driver

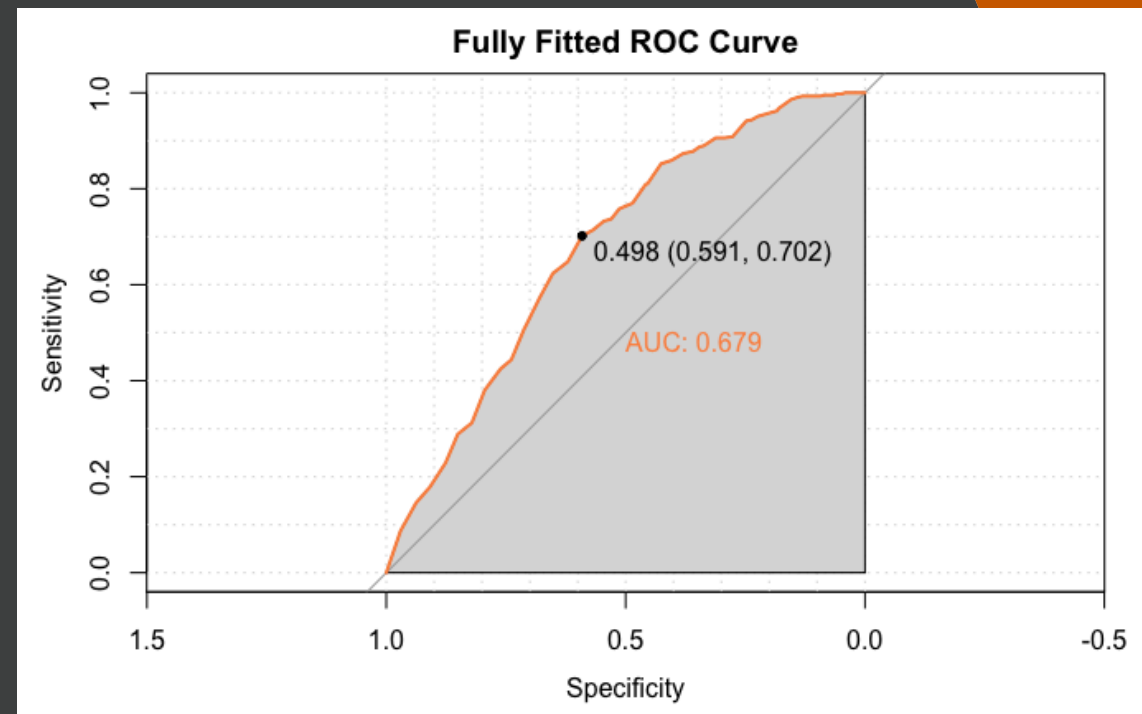


Figure 9: ROC Curve for Fully Fitted Model

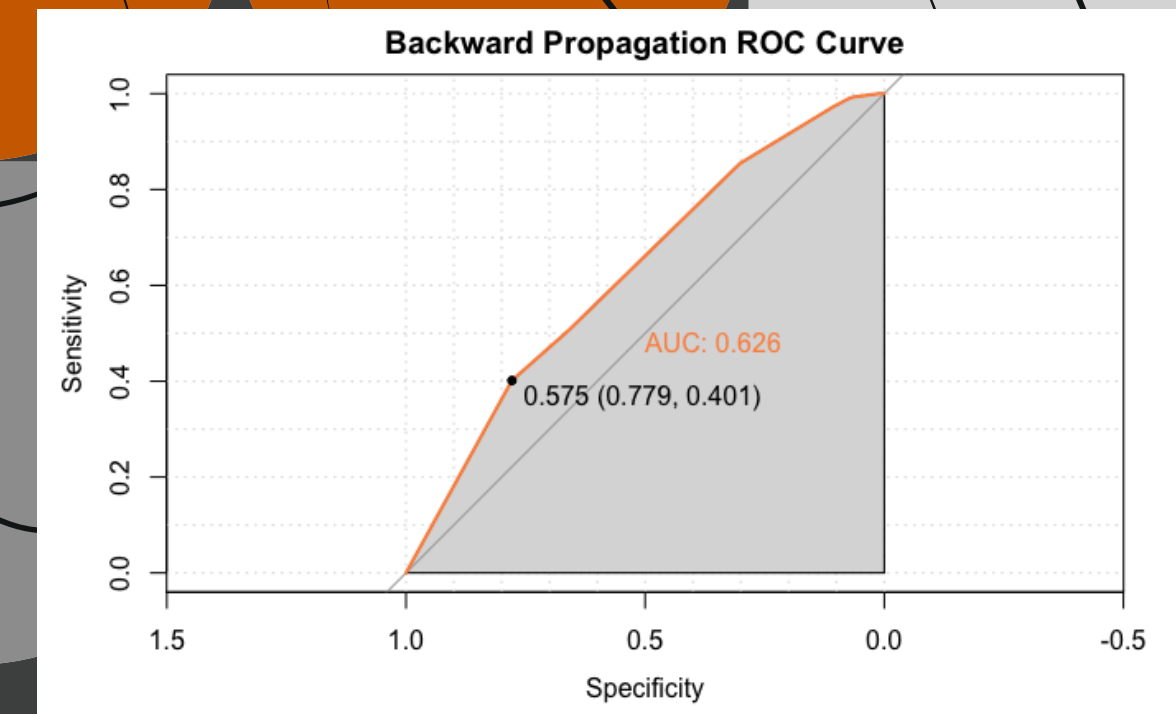


Figure 10: ROC Curve for Features Selected through Backward Propagation

Fully Fitted Model Confusion Matrix			Backward Propagation Model Confusion Matrix		
Models	True Non-Fatal	True Fatal	Models	True Non-Fatal	True Fatal
Predicted Non-Fatal	3,053	790	Predicted Non-Fatal	2,992	851
Predicted Fatal	264	162	Predicted Fatal	255	171

Model Accuracy Metrics		
	Full Model	Backward Prop
Balanced Accuracy	59.00%	58.74%
Sensitivity (Fatal)	38.03%	40.14%
Kappa	0.113	0.111

Conclusion

Random Forest Comparison by Resampling Strategy

Models	Accuracy	Sensitivity	AUC	Kappa	Outcome
Logistic Regression (Fully)	59.00%	38.03%	0.679	0.1128	Lowest sensitivity, struggling to detect fatal crashes accurately
KNN	66.20%	74.20%	0.713	0.126	Strongest overall, highest sensitivity and AUC, effective at detecting fatal crashes
Upsampled Random Forest	62.20%	67.84%	0.701	0.1226	Balanced model, AUC slightly lower but still strong at detecting fatalities

Top Predictors

- Female Driver
- Overseas Licensed Driver
- Provisional Driver

Domain Implication

- Target Overseas licensed and unlicensed/provisional drivers in policy making.
 - Stricter Requirements
 - Education

Future Work

- Expand Dataset
 - Add contextual features such as road conditions and time of day
- Explore more advanced models
 - XGBoost
 - Neural Networks

Reference List



- Li, Y., Yang, Z., Xing, L., Yuan, C., Liu, F., Wu, D., & Yang, H. (2023). Crash injury severity prediction considering data imbalance: A Wasserstein generative adversarial network with gradient penalty approach. *Accident Analysis & Prevention*, 192, 107271. <https://doi.org/10.1016/j.aap.2023.107271>
- National crash dashboard*. (2024, September 18).
<https://datahub.roadsafety.gov.au/progress-reporting/national-crash-dashboard>
- Soltani, A., Afshari, S., & Amiri, M. A. (2025). Human cost, machine insight: A data-driven analysis of Australian road crashes. *Case Studies on Transport Policy*, 20, 101440. <https://doi.org/10.1016/j.cstp.2025.101440>
- Thompson, J. P., Baldock, M. R. J., & Dutschke, J. K. (2018). Trends in the crash involvement of older drivers in Australia. *Accident Analysis & Prevention*, 117, 262–269. <https://doi.org/10.1016/j.aap.2018.04.027>
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